$\lceil 1.1$

$$f(\beta) = \|y - X\beta\|^2$$

$$= (y - X\beta)^T (y - X\beta)$$

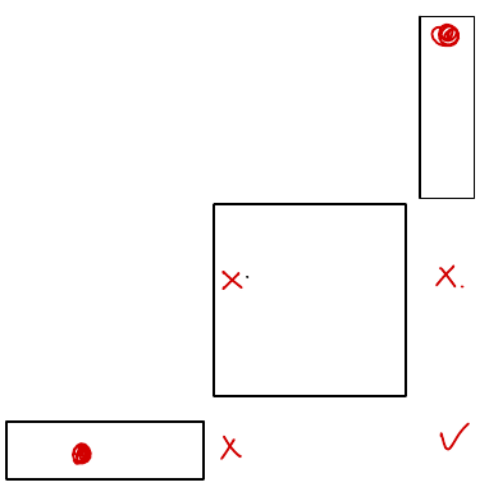
$$\nabla_\beta f(\beta) = \nabla_\beta (y^T - \beta^T X^T)(y - X\beta)$$

$$= \nabla_\beta \cancel{y^T y} - \underbrace{y^T X\beta}_{X^T y} - \underbrace{\beta^T X^T y}_{X^T y} + \underbrace{\beta^T X^T X\beta}_{2X^T X\beta}$$

$$= 2X^T X\beta - 2X^T y = 0 \quad (\text{극점})$$

then $X^T y = X^T X\beta$

직관.



$$\nabla_\beta \beta^T A \beta = \nabla_\beta \sum_{i,j} \beta_i A_{ij} \beta_j$$

$$= (A + A^T)\beta$$

1.2

- likelihood $\mathcal{L}(x_1, x_2, \dots, x_n \mid \theta)$

$$= p(x_1 \mid \theta)\, p(x_2 \mid \theta) \cdots p(x_n \mid \theta)$$

- MLE

Max likelihood estimate

$p(X \mid \theta)$를 높이는 θ를 찾자!

Bernoulli random variable

$$p\{x_1 = 1 \mid \theta\} = \theta$$

$$p\{x_1 = 0 \mid \theta\} = 1 - \theta$$

$$p(x_i \mid \theta) = \theta^{x_i} \cdot (1-\theta)^{1-x_i} \xrightarrow{\log} x_i \log\theta + (1-x_i)\log(1-\theta)$$

$$p(x \mid \theta) = \theta^{\sum x_i} \cdot (1-\theta)^{n-\sum x_i} \longrightarrow \sum x_i \log\theta + (n - \sum x_i)\log(1-\theta)$$

θ에 대한 미분!

$$\sum x_i \cdot \frac{1}{\theta} + (n - \sum x_i)\frac{-1}{1-\theta} = 0$$

$$(1-\theta)\sum x_i = (n - \sum x_i)\theta$$

$$\theta = \frac{\sum x_i}{n}$$

$p(\theta)$ prior

$p(data|\theta)$ likelyhood

$p(data)$ evidence

$p(\theta|data)$ posterior

예시 θ : 감기

data : 기침.

$$\boxed{p(\theta|data) = \frac{p(data|\theta)\,p(\theta)}{p(data)} \propto p(data|\theta)\cdot p(\theta)}$$

MAP : $p(\theta|data)$ 를 높이는 θ를 찾기.

Max A Posterior Estimate

Bernoulli

let $p(\theta) =$

$P\{x_1=1|\theta\} = \theta$

$P\{x_1=0|\theta\} = 1-\theta$.

$$\boxed{\begin{aligned} p(x_i|\theta) &= \theta^{x_i}\cdot(1-\theta)^{1-x_i} \\ p(x|\theta) &= \theta^{\sum x_i}\cdot(1-\theta)^{n-\sum x_i} \end{aligned}}$$

$p(\theta|data) \propto p(\theta)\cdot p(data|\theta)$

$p(\theta|data) \propto \theta^{\sum x_i}(1-\theta)^{n-\sum x_i}$ $(0<\theta<1)$

$$p(\theta|data) = \left(\frac{\Gamma(n+2)}{\Gamma(\sum x_i+1)\,\Gamma(n-\sum x_i+1)}\cdot\theta^{\sum x_i}(1-\theta)^{n-\sum x_i}\right) \text{ in } 0<\theta<1$$

$\Gamma(n) = (n-1)!$

$\Rightarrow$ $p(\theta)=1$ 라는 건 θ에 대한 정보가 없음을 의미한다.
$(0<\theta<1)$

$\Rightarrow$ MLE = MAP

$X \sim Beta(\alpha, \beta)$

pdf $\quad f(x) = (\text{constant}) \cdot x^{\alpha-1} \cdot (1-x)^{\beta-1}$ $\quad 0 \le x \le 1$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\cdot\Gamma(\beta)} x^{\alpha-1} \cdot (1-x)^{\beta-1}$$ $\quad \Gamma(n) = (n-1)!$ $\quad$ (오른쪽은 아니다.)

Bernoulli Distribution. $\quad$ H $\quad$ T

7 $\quad$ 3

MLE $\quad$ · $p(\text{data}|\theta) = \theta^7 (1-\theta)^3 \longrightarrow \theta_{MLE} = 0.7$

MAP $\quad$ · let $\theta \sim Beta(2,2)$

$$p(\theta) = \frac{\Gamma(4)}{\Gamma(2)\cdot\Gamma(2)} \theta(1-\theta)$$

$$= \frac{3!}{1!\cdot 1!} \theta(1-\theta) \qquad 0 \le \theta \le 1$$

· $p(\theta|\text{data}) \propto p(\theta) \cdot p(\text{data}|\theta)$

$$\propto \theta^8 (1-\theta)^8 \longrightarrow MAP = 0.666\ldots$$

Amazon. $\quad$ MAP

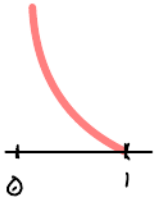
	data ⊕	data ⊖		assume
Seller ①	90	10	$\longrightarrow \theta_1$	$p(\theta_1) = 1 \quad 0 < \theta_1 < 1$
Seller ②	2	0	$\longrightarrow \theta_2$.	$p(\theta_2) = 1 \quad 0 < \theta_2 < 1$

$p(\theta_1|\text{data}) \propto \theta_1^{90} (1-\theta_1)^{10}$

$p(\theta_2|\text{data}) \propto \theta_2^{2}$.

$$p(\theta_1 > \theta_2|\text{data}) = \int_0^1\int_0^1 I\{\theta_1 > \theta_2\}\, Beta(\theta_1|91,11)\, Beta(\theta_2|3,1)\, d\theta_1 d\theta_2 \approx 71\%$$

- $h(x) = -\log p(x)$



- indep x, y : $h(x,y) = h(x) + h(y)$

- entropy : $H(X) = E[-\log p(X)]$

① $H(X) \geq 0$

② $H(X) = 0$ if $\exists x \; p(x) = 1$

$$H(X,Y) = H(X) + H(Y|X)$$

$$-\log p(X,Y) = -\log p(X) - \log \frac{p(X,Y)}{p(X)}$$

- $D_{KL}(p\|q) = \sum_x p(x) \log \frac{p(x)}{q(x)}$

- symmetric (X)
- triangular inequality (X)

example. Bernoulli $\begin{cases} p \Rightarrow r \\ q \Rightarrow s \end{cases}$

$$D_{KL}(p\|q) = (1-r) \log \frac{1-r}{1-s} + (r) \log \frac{r}{s}$$

$D_{KL}(p\|q) \geq 0$ $\quad \left[D_{KL}(p\|q) = 0 \text{ iff } \forall x \; p(x) = q(x) \right]$

Jenson's Inequality : f: convex.

$E[f(x)] \geq f(E(x))$

equality holds when $f(x) = \text{constant}$.

pf. $D_{KL}(p\|q)$

$$= E_{p(x)}\left(-\log \frac{q(x)}{p(x)}\right)$$

$$\geq -\log E_{p(x)}\left[\frac{q(x)}{p(x)}\right]$$

$$= -\log \sum_x \cancel{p(x)} \cdot \frac{q(x)}{\cancel{p(x)}} = 0$$

특수 조건 $\quad \text{const} = \frac{q(x)}{p(x)} = 1$

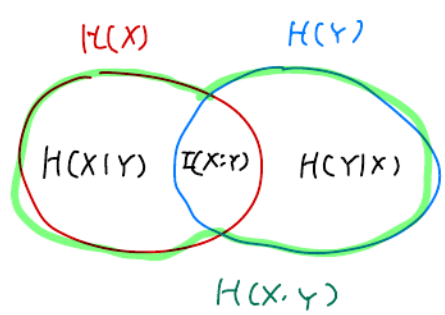
$\begin{cases} D_{KL}(p\|q) \Downarrow : \text{모든 영역} \\ D_{KL}(q\|p) \Downarrow : \text{동우리} \end{cases}$

$$I(X:Y) = D_{KL}(p(x,y) \| p(x)p(y))$$

$$= \sum p(x,y) \cdot \log \frac{p(x,y)}{p(x)p(y)}$$

$$= H(X) + H(Y) - H(X,Y)$$

$$= H(X) - H(X|Y)$$



- Cross Entropy $H_p(q) = E_p[-\log q(X)]$

$$= -\sum_x p(x) \log q(x)$$

- $D_{KL}(p\|q) = \sum_x p(x) \log \frac{\overset{H(p)}{p(x)}}{\underset{H_p(q)}{q(x)}}$

$$= H_p(q) - \cancel{H(p)} \geq 0$$

- p가 실제, q가 예측모델

$D_{KL}(p\|q)$ 최소인 $H_p(q)$ 사용 가능

- let $y = \underbrace{\beta_0 \cdot 1 + \beta_1 \cdot x + \beta_2 x + \cdots + \beta_m x_m}_{\langle x_i, \beta \rangle} + \varepsilon$ $\sim N(0, \sigma^2)$

- $Y_i | X_i, \beta \sim N(\langle X_i, \beta \rangle, \sigma^2)$

- $P(Y_i | X_i, \beta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \langle x_i, \beta \rangle)^2}{2\sigma^2}\right) \xrightarrow{\log} \text{상수} - \frac{(y_i - \langle x_i, \beta \rangle)^2}{2\sigma^2}$

- $P(Y | X, \beta) \leftarrow$ MLE

$$= P(y_1, y_2, \ldots, y_n \mid x_1, \ldots, x_n, \beta)$$

$$= P(y_1 | x_1, \beta) \cdot P(y_2 | x_2, \beta) \cdots P(y_n | x_n, \beta)$$

Max Likelyhood Estimate = Least Square Estimate.

$\xrightarrow{\log}$ 상수 $- \sum_i \frac{(y_i - \langle x_i, \beta \rangle)^2}{2\sigma^2} \leftarrow$ Least Square Estimate